Vitalii Belov

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AI/ML Engineer with a focus on NLP, LLMs, Graph ML, reinforcement learning, and multi-agent systems. Interested in building scalable, user-facing intelligent systems and agentic products. Open to relocation.

Summary

AI/ML Engineer with 3+ years of experience across applied research and production ML. Specialized in end-to-end **LLM/RAG** and **Graph RAG** systems, **multi-agent systems (MAS)**, and **RL environments for LLM post-training**. Strong **MLOps** practices (Docker, CI/CD, monitoring) and solid engineering background in Python/C++. Seeking roles in NLP/LLMs, agentic systems, graph-based ML, and low-latency AI systems.

Core Skills

LLMs, RAG & Agents: Retrieval, Prompt Engineering, Graph RAG, Agent Topologies, Tool Integration, Offline Evaluation, Agent Orchestration.
RL & LLM Post-Training: RL environments, reward-oriented optimization pipelines, evaluation loops, training stability.
Graph ML & MAS: Graph-based representations, cross-document linking, multi-agent communication design/optimization.
ML & Data: Supervised/Unsupervised Learning, Feature Engineering, Metrics (F1, Precision/Recall).
MLOps & Deployment: Docker, CI/CD (GitHub Actions), Experiment Tracking (MLflow), Monitoring, Basic Kubernetes.
Programming: Python (PyTorch, NumPy, Pandas, scikit-learn, SciPy, Matplotlib), C++, SQL.
Infra/Tools: Kafka, Hadoop, Git.
Languages: Russian (Native), English (Fluent), Mandarin Chinese (Intermediate, HSK 4), Japanese (N3).

Professional Experience

Anthropic — Data Team *ML Engineer / Research Engineer (RL for LLM Training)* Jan 2026 – Mar 2026

- Worked on **RL infrastructure and environments** for post-training and optimization of Anthropic language models.
- Configured and improved **reinforcement learning environments** used in training loops for alignment and model quality improvement.
- Contributed to the stability, scalability, and evaluation of RL training pipelines for frontier LLM systems.
- Focused on environment setup, reward-oriented optimization workflows, and training/evaluation reliability.

Sberbank — Graph DS / MAS *Data Scientist / AI Researcher (Graphs, MAS, LLMs)* Aug 2025 – Present

- Conduct research in **graph models** and **multi-agent systems** for LLM-based products: agent communication topologies, interaction optimization, and tool integration.
- Built **multi-agent systems for internal teams across the bank**, including domain-specific assistants such as HR analytics bots and similar internal AI products.
- Built a **multi-agent agent-generation system** using **Qwen 4B** that outperformed alternative implementations while being significantly cheaper.
- Developed an in-house framework as an alternative to **LangGraph** for MAS with **dynamic topology**, decentralization, and automatic task-specific agent-graph generation.
- Stack: LangChain, LangGraph, Python, PyTorch, Docker, NLP, Graph ML, Multi-Agent Systems.

Sberbank — IDP (Intelligent Document Processing) *Data Scientist / ML Engineer (NLP)* Apr 2025 – Jul 2025

- Developed an internal **RAG** and **Data Enrichment** service for document understanding, extraction, and validation.
- Improved answer quality by **+4 points on average** and by **+12** on meta-questions; added **meta-chunk** support.
- Implemented entity extraction and classification in the document processing pipeline; designed **Graph RAG** architecture for richer context.
- Productionized components with Docker and basic CI/CD practices; contributed to stability/latency monitoring.

Applied Research Laboratory, VNIIA (Rosatom) *Python/C++ Developer & Algorithm Engineer* Sep 2022 – Apr 2025

- Developed and optimized classical and **quantum** algorithms in Python/C++; supported research projects (NDA).
- Built reproducible data processing and evaluation pipelines; applied ML/NLP for recommendation-related tasks in research settings.
- Contributed to scientific publications (see Publications).

Tinkoff — Project Internship *Analyst* Jun 2022 – Jul 2022

- Performed data analysis and reporting; optimized SQL queries; delivered a complete case solution with the team.

Personal Projects

Trading Bot (Tinkoff Invest API) *Python, APIs* Jan 2024 – Feb 2024

- Built a personal trading bot integrating with Tinkoff API; implemented data ingestion, strategy execution, and basic monitoring/logging.

Dota 2 Match Outcome Predictor (DraftEdge) *Python, scikit-learn, Pandas* Jan 2024 – Feb 2024

- Trained an ML model to predict professional match outcomes based on draft/player statistics; used feature engineering and cross-validation.

Game Lore AI Assistant *Python, LLMs, RAG, Vector DBs* 2024

- Designed a RAG-based assistant for context-dependent Q&A over a large corpus of game narrative documentation.
- Built ingestion/chunking and embedding pipeline; tuned prompts for coherence and factual grounding.

Competitions & Hackathons

- Stock trading hackathon (Spectral::Technologies).
- Screening hackathon for Kontur (Data Scientist track).
- Screening hackathon for Sberbank team (NLP track).
- Gazprom ML course capstone hackathon.

Publications

- *LLM Agents, Retrieval-Augmented Systems, Low-Latency AI Systems.* Submitted to a conference; currently under review. Code and dataset release: frontier-ai/llm-agent-factory.
- *gMAS: A Framework for Dynamic Graph-Based Multi-Agent LLM Orchestration (Demo).* Submitted to a conference; currently under review.
- *Measurement induced phase transition in the central spin model: second Rényi entropy in dual space approach.*
- *Dynamical quantum Ansatz tree approach for the heat equation.*

Education

Moscow Institute of Physics and Technology (MIPT) 2024 – 2026
M.Sc. in Applied Mathematics and Computer Science (Problems of Modern Mathematics)

National Research University Higher School of Economics (HSE) 2020 – 2024
B.Sc. in Physics, specialization: Quantum Computing

Additional Courses: Deep Learning School (2024), Yandex Algorithm Training (2024), Samsung Computer Vision Course (2024), Vega Institute (Financial Math, Monte Carlo, Game Theory) (2025), Yandex Handbooks (2024–2025).